

How-To Guide: Maintenance Planner

Role: Maintenance Planner **Modules:** Module 2 (Work Request Processing), Module 3 (Backlog & Scheduling), Module 4 (Strategy) **Pages:** Page 9 (Work Requests), Page 10 (Planner Assistant), Page 11 (Backlog & Schedule), Page 12 (Weekly Scheduling), Page 4 (Strategy -- Tasks/Work Packages)

Role Description

Maintenance Planners are responsible for processing incoming work requests, reviewing AI-generated recommendations, managing the maintenance backlog, creating weekly maintenance programs, and building work packages. They act as the bridge between field observations and scheduled maintenance execution.

Prerequisites

Requirement	Details
API server running	<code>uvicorn api.main:app --reload</code> on <code>http://localhost:8000</code>
Streamlit app running	<code>streamlit run streamlit_app/Home.py</code> on <code>http://localhost:8501</code>
Database seeded	Plants, equipment hierarchy, and sample data loaded (Page 7 seed)
Work requests exist	At least one field capture submitted (Page 8) generating work requests
Planner permissions	Access to approve, modify, escalate, or defer work requests

User Story MP-01: Process Incoming Work Requests

Goal: Review incoming work requests from field captures, validate AI classifications, and approve or reject them.

Step-by-Step Instructions

1. **Navigate to Page 9 -- Work Requests** - Open the Streamlit sidebar and click **Work Requests**.

2. **Filter by Status** - Use the **Filter by Status** dropdown and select **DRAFT** or **PENDING_VALIDATION** to see items requiring your attention.
3. **Review the Work Request Queue** - The page shows a count header: "X Work Requests Found". - Each work request is displayed as an expandable card showing:
 - Request ID (truncated)
 - Equipment tag
 - Status badge (color-coded)
4. **Expand and Review Each Work Request** - Click on a work request expander to see details:
 - **Equipment**: Matched equipment tag
 - **Confidence**: AI matching confidence percentage
 - **AI Priority**: Suggested priority (e.g., **2_URGENT**)
 - **WO Type**: Work order type (CM, PM, etc.)
 - **Duration**: Estimated hours
 - **Specialties**: Required maintenance disciplines
 - **Created**: Submission timestamp
5. **Take Action** - For each **DRAFT** or **PENDING_VALIDATION** work request, use the three action buttons:
 - **Approve**: Accept the work request for further processing by the planner.
 - **Reject**: Reject the work request with a reason.
 - **Classify**: Re-trigger AI classification to refresh the analysis.
 - The page automatically reloads after each action.

API Endpoints Called

```
GET /api/v1/work-requests/?status=PENDING_VALIDATION
PUT /api/v1/work-requests/{request_id}/validate { "action": "APPROVE" }
POST /api/v1/work-requests/{request_id}/classify
```

Expected Output

- Work request status changes from **DRAFT** / **PENDING_VALIDATION** to **VALIDATED** or **REJECTED** .
- Success message: "Work request {id} approved" or "Work request {id} rejected".
- The page refreshes to show updated status badges.

Error Handling

Issue	Solution
"Could not load work requests"	Verify the API is running. Check the database has been seeded.
Approve button returns error	The work request may have already been processed. Refresh the page.
AI classification seems incorrect	Click Classify to re-run AI analysis with updated context.

User Story MP-02: Review Planner Recommendation

Goal: Use the AI-powered Planner Assistant to generate a detailed recommendation for a specific work request, including resource analysis, scheduling suggestions, and risk assessment.

Step-by-Step Instructions

1. **Navigate to Page 10 -- Planner Assistant** - Open the Streamlit sidebar and click **Planner Assistant**.
2. **Select a Work Request** - Use the dropdown to select a work request from the list. - The dropdown shows: Request ID | Equipment Tag | Status. - Three metric cards display: Equipment, Status, and AI Priority.
3. **Generate AI Recommendation** - Click the **Generate Recommendation** button (blue primary button). - Wait for the AI engine to analyze the work request.
4. **Review the Recommendation** - Once generated, the recommendation displays:
 - **Suggested Action:** APPROVE, MODIFY, ESCALATE, or DEFER
 - **Confidence:** AI confidence in the recommendation
 - **Risk Level:** Overall risk assessment
5. **Explore the Three Analysis Tabs**

Resources Tab: - Workforce availability by specialty (e.g., "MECHANICAL: 3 available") - Materials status (all available or missing items listed)

Scheduling Tab: - Recommended date and shift - Scheduling reasoning - Any conflicts with existing schedule - Groupable backlog items (items that can be combined)

Risk Tab: - Risk level assessment - Individual risk factors - Mitigation recommendation

6. **Apply Planner Action** - Below the recommendation, four action buttons are displayed:
 - **APPROVE:** Accept the recommendation and move to scheduling.
 - **MODIFY:** Accept with modifications (planner adjusts parameters).
 - **ESCALATE:** Escalate to management for higher-level decision.

- **DEFER:** Defer the work to a later period.
- Click the appropriate action button.

API Endpoints Called

```
POST /api/v1/planner/{work_request_id}/recommend
PUT /api/v1/planner/recommendations/{recommendation_id}/action
{ "action": "APPROVE" }
```

Expected Output

Recommendation response:

```
{
  "recommendation_id": "uuid-...",
  "planner_action": "APPROVE",
  "ai_confidence": 0.87,
  "risk_level": "MEDIUM",
  "resource_analysis": {
    "workforce_available": [
      { "specialty": "MECHANICAL", "technicians_available": 3 }
    ],
    "materials_status": { "all_available": true }
  },
  "scheduling_suggestion": {
    "recommended_date": "2025-03-15",
    "recommended_shift": "DAY",
    "reasoning": "Equipment criticality is HIGH...",
    "conflicts": [],
    "groupable_with": ["BL-002", "BL-005"]
  },
  "risk_assessment": {
    "risk_level": "MEDIUM",
    "risk_factors": ["Equipment criticality AA", "No redundancy"],
    "recommendation": "Schedule within 7 days"
  }
}
```

Error Handling

Issue	Solution
"No work requests available"	Submit field captures on Page 8 first. Ensure work requests exist.
Recommendation generation fails	Check API logs. Ensure the work request has valid equipment data.
Action application fails	The recommendation may have already been actioned. Refresh the page.

User Story MP-03: Manage Backlog

Goal: View, filter, and optimize the maintenance backlog to prioritize and group work items.

Step-by-Step Instructions

1. **Navigate to Page 11 -- Backlog & Schedule** - Open the Streamlit sidebar and click **Backlog & Schedule**. - The page has three tabs: **Backlog**, **Optimize**, and **Schedule**.
2. **Review Current Backlog (Backlog Tab) - Filter by Status:** Select from:
 - **AWAITING_MATERIALS**
 - **AWAITING_SHUTDOWN**
 - **AWAITING_APPROVAL**
 - **SCHEDULED**
 - **Filter by Priority:** Select from:
 - **1_EMERGENCY**
 - **2_URGENT**
 - **3_NORMAL**
 - **4_PLANNED**
 - View summary metrics:
 - **Total Items:** Count of backlog items
 - **Materials Ready:** Items with all materials available
 - **Total Hours:** Sum of estimated work hours
 - A priority distribution pie chart visualizes the breakdown.
 - The full backlog data table is shown below.
3. **Run Backlog Optimization (Optimize Tab)** - Switch to the **Optimize** tab. - Enter the **Plant ID** (default: **OCP-JFC1**). - Set the **Planning Horizon** using the slider (7 to 90 days, default: 30). - Click **Run Optimization** (blue primary button). - Review optimization results:
 - **Total Items:** Total backlog items analyzed
 - **Schedulable Now:** Items ready to schedule
 - **Blocked:** Items with unresolved dependencies
 - **Work Packages:** Suggested work package groupings
 - A stratification chart shows the backlog breakdown.
 - Click **Approve Schedule** to confirm the optimized schedule.
4. **View Latest Schedule (Schedule Tab)** - Switch to the **Schedule** tab. - View the approved schedule:
 - Status and period date range
 - Utilization chart showing resource allocation
 - Work packages included in the schedule

- Alerts for overdue items, priority escalations, or material delays

API Endpoints Called

```
GET /api/v1/backlog/?status=AWAITING_MATERIALS&priority=2_URGENT
POST /api/v1/backlog/optimize
    { "plant_id": "OCP-JFC1", "period_days": 30 }
PUT /api/v1/backlog/optimizations/{optimization_id}/approve
GET /api/v1/backlog/schedule
```

Expected Output

Optimization result:

```
{
  "optimization_id": "uuid-...",
  "total_items": 25,
  "schedulable_now": 18,
  "blocked": 7,
  "work_packages": 5,
  "stratification": { "emergency": 2, "urgent": 5, "normal": 11, "planned": 7 }
}
```

Error Handling

Issue	Solution
"No backlog items"	Approved work requests must be added to the backlog. Process work requests on Pages 9-10 first.
Optimization returns 0 schedulable	Check material readiness and equipment availability. Adjust the planning horizon.
Schedule approval fails	Ensure the optimization has not already been approved.

User Story MP-04: Create Weekly Maintenance Program

Goal: Create and manage weekly maintenance programs that allocate work packages to specific weeks, track resources, and generate Gantt charts.

Step-by-Step Instructions

1. **Navigate to Page 12 -- Weekly Scheduling** - Open the Streamlit sidebar and click **Weekly Scheduling**. - The page has three tabs: **Programs**, **Resources**, and **Gantt**.

2. **Create a New Program (Programs Tab)** - Enter the **Plant ID** (default: `OCP-JFC1`). - Set the **Week Number** (1--53). - Set the **Year** (default: 2025). - Click **Create Program** (blue primary button). - A success message shows the created program ID.
3. **Review Existing Programs** - Below the creation form, existing programs are listed with:
 - Program ID (truncated)
 - Week/Year
 - Status
 - Total hours
 - For `DRAFT` programs, a **Finalize** button is available.
 - Click **Finalize** to lock the program for execution.
4. **Check Resource Utilization (Resources Tab)** - Switch to the **Resources** tab. - View resource data for the most recently created program:
 - **Total Hours**: Sum of planned work hours
 - **Work Packages**: Number of work packages assigned
 - **Conflicts**: Number of resource conflicts detected
 - A utilization chart shows daily resource allocation percentages.
 - Any scheduling conflicts are displayed as warnings.
5. **View Gantt Chart (Gantt Tab)** - Switch to the **Gantt** tab. - The Gantt chart shows work packages plotted on a timeline. - Use the **Download Excel** button to export the Gantt data (via the API export endpoint).

API Endpoints Called

```
POST /api/v1/scheduling/programs
  { "plant_id": "OCP-JFC1", "week_number": 12, "year": 2025 }
GET  /api/v1/scheduling/programs?plant_id=OCP-JFC1
GET  /api/v1/scheduling/programs/{program_id}
PUT  /api/v1/scheduling/programs/{program_id}/finalize
GET  /api/v1/scheduling/programs/{program_id}/gantt
GET  /api/v1/scheduling/programs/{program_id}/gantt/export
```

Expected Output

Program creation response:

```
{
  "program_id": "uuid-...",
  "plant_id": "OCP-JFC1",
  "week_number": 12,
  "year": 2025,
  "status": "DRAFT",
  "total_hours": 156.5,
  "work_packages": [...],
  "resource_slots": [...],
  "conflicts": []
}
```

Error Handling

Issue	Solution
"Error creating program"	Ensure backlog items exist and have been optimized. Check API logs.
No work packages in program	Run backlog optimization first (Page 11, Optimize tab).
Finalize fails	Resolve all resource conflicts shown in the Resources tab first.
Empty Gantt chart	Create a program first. Ensure it contains work packages.

User Story MP-05: Build Work Packages

Goal: View, validate, and approve maintenance work packages that group related tasks together.

Step-by-Step Instructions

- Navigate to Page 4 -- Strategy** - Open the Streamlit sidebar and click **Strategy**. - The page has three tabs: **Tasks**, **Work Packages**, and **Naming Validation**.
- Review Maintenance Tasks (Tasks Tab)** - View the list of all maintenance tasks with columns:
 - Task ID, Name, Task Type, Status
 - This provides context for which tasks are available for packaging.
- Review Work Packages (Work Packages Tab)** - Switch to the **Work Packages** tab. - View all work packages with columns:
 - Work Package ID, Name, Code, Status
 - For **DRAFT** work packages:
 - Select the work package from the dropdown.
 - Click **Approve Work Package** to change status to **APPROVED**.

4. **Validate Naming Conventions (Naming Tab)** - Switch to the **Naming Validation** tab. - **Task Name Validation:**

- Enter a task name (e.g., `"INSPECT_BEARING_CONDITION_SAG_ML_001"`).
- Select the task type (INSPECT, CHECK, TEST, LUBRICATE, CLEAN, REPLACE, REPAIR, CALIBRATE).
- Click **Validate Task Name**.
- Green success message if valid; warnings list any naming issues.
- **Work Package Name Validation:**
- Enter a work package name.
- Click **Validate WP Name**.

API Endpoints Called

```
GET /api/v1/tasks/
GET /api/v1/work-packages/
PUT /api/v1/work-packages/{wp_id}/approve
POST /api/v1/tasks/validate-name { "name": "...", "task_type": "INSPECT" }
POST /api/v1/tasks/validate-wp-name { "name": "..." }
```

Expected Output

Work package approval:

```
{
  "work_package_id": "uuid-...",
  "status": "APPROVED"
}
```

Task name validation:

```
{
  "valid": true,
  "issues": []
}
```

Error Handling

Issue	Solution
"Cannot connect to API"	Start the API server. Verify the database is seeded.
No draft work packages available	Work packages must be created first (either via seed data or from FMEA/strategy flow).
Naming validation returns issues	Follow the suggested corrections in the warning messages.
Approval fails with 409	The work package may already be approved or in an invalid state.

Quick Reference: Planner Workflow

